nature portfolio

# Peer Review File

# Enhanced diagnosis of multi-drug-resistant microbes using group association modeling and machine learning

Corresponding Author: Professor Tony Hu

**This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.**

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
In this work, Saliba and colleagues implement a novel Group Association Model (GAM) to identify correct single gene associations with their corresponding drug resistance while avoiding common and misleading cross-associations due the presence of overlapping drug resistances in clinical Mtb strains. Their approach consisted in stratifying the dataset into groups of isolates sharing the same antibiogram. They show that their approach was more specific that standard GWAS LMM in detecting AMR genes, while not detecting cross-associations. They next showed that ML models built on genetic variants from GAM detected genes outperformed the sensitivity of drug resistance predictions made by the WHO mutation catalogue.

Major comments:

- About the power of GAM compared to standard GWAS LMM method. The authors found that “No significant associations were detected for the four new or repurposed drugs (bedaquiline, clofazimine, delamanid, and linezolid), likely due to scarcity of groups/isolates resistant to them.” This makes we wonder what the power loss attributable to the groupings required to perform GAM. While the authors show, and it is widely recognised, that GWAS LMM will detect multiple false-positive gene associations for most drugs, due to the common co-occurrence of drug resistances, this approach may still be more powered to detect true associated genes for new and repurposed drugs. The authors should compare the power of detecting true associations of GAM compared to standard LMM approaches.

- The concept of validating GAM or any other genotype-phenotype association method on antibiotic resistance datasets. The authors state that “To validate GAM performance, we next employed it to analyze a Staphylococcus aureus” dataset. However, pretty much most regression-based and ML approaches are expected to identify SNP resistance-associated and acquired AMR genes in such datasets, given the strong size effects of AMR genetic determinants in bacteria. As stated above, the authors should provide comparisons of methods in terms of power and false positive rates, and overall number of know AMR genetic determinants identified.

- The authors should provide the raw GAM and LMM results per drug, that is, the table of all associated genetic variants and genes, ordered increasingly by p-value, not only the known AMR determinants presented in Supplementary Table 1 and 2. For example, it is unlikely that plasmid-borne AMR genes will be found associated on their own, but most likely along with other plasmid carried genes.

- The section comparing GAM results to the WHO Mtb mutations catalogue (lines 174-221) is rather hard to follow. The authors show that training an ML model on all genetic variants from GAM detected genes outperforms the sensitivity of drug resistance predictions, when compared to the predictions made by GAM variants only, and the WHO 21 and 23 mutations catalogues (Figure 4d). However, the authors should also show the PPV and Specificity of the GAM+ML model in Figure 4b and 4c.

- The section on “Impact of sample size and data completeness on GAM performance” is very relevant and informative. As mentioned above, reporting the power and false positive rates with varying samples sizes of the GAM model, compared to commonly used LMM, will be informative too.

- The results presented in Figure 6 will be more informative if presented in terms of sensitivity and specificity rather than overall accuracy.

- Data and code availability: the GitHub repository github.com/jsaliba1/GAM does not exist, so the existence of code could not be reviewed. The section “Availability of data and materials” should also include the source of S. aureus genomes and linked AST data, as well as the genome accessions of the newly sequenced 428 DS3 isolates from China with linked DST profiles.

Minor comments

Introduction

Line 46. Spell out Mycobacterium tuberculosis (Mtb) in the first instance in the Introduction.

Lines 56-58. Drug resistance can arise on a background of pre-existing drug resistance. This is due to the nature of TB treatment being a combination therapy (multiple drugs used in regimens), and regimens being standardised for increasingly resistant strains (regimens for MDR and XDR strains). Please state and extend about how these treatment choices result in these overlapping resistances in clinical Mtb strains.

Line 271. Correct misspelt “ethanamide” drug name.

(Remarks on code availability)
The GitHub repository github.com/jsaliba1/GAM does not exist, so the existence of code could not be reviewed.

Reviewer #2

(Remarks to the Author)
The authors present an overall interesting approach of predicting multi-drug-resistance for Mtb samples from the CRyPTIC database without additional prior knowledge or expert rules. However, the detailed procedure seems unclear and intermediate results, specifically also in code / table format are missing for proper evaluation.

MAJOR COMMENTS

1) Code and Method Clarity: Lack of access to the code and insufficient methodological details hinder a full evaluation, especially for GAM procedures and sample size/missing data handling.

2) GAM Output and Validation: The selection criteria for resistance groups, exact GAM output, and its performance evaluation (particularly against the WHO catalog) need clarification, ideally with validation on independent or held-out data.

3) Supplemental Data and Consistency: Complete tables of quantitative results, especially significant variants for GAM groups, should be provided as supplemental data. Consistent terminology for drug names (e.g., Rifampin vs. Rifampicin) is also needed.

4) GAM Generalization and Horizontal Gene Transfer: The manuscript lacks clarity on GAM’s generalization for S. aureus and the limitations in detecting horizontal gene transfer, requiring further explanation.

For more details, see minor comments below.

MINOR COMMENTS

1) The code repository was not available to the reviewer, therefore it couldn’t be used to evaluate procedures with unclear description in the methods.

2) How were the different resistance groups evaluated in the GAM selected? All possible combinations that were observed?

3) Association results of all variants, but specifically significant variants for all GAM groups should be supplied as Supplemental Data.

4) Similar tables of complete quantitative results should be supplied for all following analyses as well, especially also for data shown in the figures.

5) How are gene level associations evaluated in the GAM procedure? From Figure 3 A, the Supplemental Figure 2 and the methods (line 409-411), this is unclear.
It seems like the outline is switching from variant level to gene level and back to variant level.

6) What was the exact output of the GAM procedure and how was it evaluated for performance? Was the WHO catalog used as ground truth/gold standard?
Predictive performance should be evaluated on an independent dataset. At the least, performance needs to be evaluated

using unseen, held out data.

7) The description of the generalization of the GAM for S. aureus is insufficient and unclear. Specifically, what was the “global assessment” as mentioned in ”comparisons between groups and controls were substituted with a global assessment of whether mutations differed from the control group.” (line 432-433)?

8) Since basic methods are unclear, it is difficult to assess if the sample size and missing data investigations are reasonable.

9) Throughout the manuscript, Rifampin and Rifampicin are used. The authors should keep using only one name for one drug.

10) Would it be possible to clarify, why it should be challenging to detect horizontal gene transfer with the GAM, as it is based on associations and not e.g. phylogenetic trees?

11) Figure 3d: legend for the color is missing.

12) WHO catalog contents and expert ruling should be explained very briefly in the context on how that approach differs from the GAM/ML analyses.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)
Code repository was not available for review.

Version 1:

Reviewer comments:

Reviewer #2

(Remarks to the Author)
Comments to the authors:

The authors significantly improved their manuscript, making their work much more transparent and reproducible. Also, the code repository is now available and well structured, and data to run the analyses is provided.

Remaining comments

1) The full data for the GAM analysis is available, but in addition a table listing the 127 resistance groups as well as one with the assignments of the samples to the different groups would be helpful.

2) Any description of the LMM analysis is missing in the method section.

3) Abbreviations in tables (e.g. SEN_lci in Supplementary Table 3) should be explained.

4) Supplementary Table 4: Please add at least the TN numbers, potentially also PPV or accuracy, so that full performance metrics are available.

5) For reproducibility, a short summary for the most important items in the supplied data folder on Figshare (e.g. content and dimensions, what would be input file, what is a result file, …) would be very helpful.

(Remarks on code availability)
Well structured.

Reviewer #3

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career

Researchers who co-review manuscripts.

(Remarks on code availability)

**Response to Reviewer #1:**

In this work, Saliba and colleagues implement a novel Group Association Model (GAM) to identify correct single gene associations with their corresponding drug resistance while avoiding common and misleading cross-associations due the presence of overlapping drug resistances in clinical Mtb strains. Their approach consisted in stratifying the dataset into groups of isolates sharing the same antibiogram. They show that their approach was more specific that standard GWAS LMM in detecting AMR genes, while not detecting cross-associations. They next showed that ML models built on genetic variants from GAM detected genes outperformed the sensitivity of drug resistance predictions made by the WHO mutation catalogue.

We sincerely thank the reviewer for their thoughtful and detailed comments on our work, and their recognition of our approach's utility for mitigation of cross-associations caused by overlapping drug resistances in *Mtb* isolates, and its ability to outperform predictions based on the WHO mutation catalogue.

We have carefully addressed all comments and suggestions in the revised manuscript, and provide detailed point-to-point responses as follows:

**Major comments:**

1) About the power of GAM compared to standard GWAS LMM method. The authors found that "No significant associations were detected for the four new or repurposed drugs (bedaquiline, clofazimine, delamanid, and linezolid), likely due to scarcity of groups/isolates resistant to them." This makes we wonder what the power loss attributable to the groupings required to perform GAM. While the authors show, and it is widely recognised, that GWAS LMM will detect multiple false-positive gene associations for most drugs, due to the common co-occurrence of drug resistances, this approach may still be more powered to detect true associated genes for new and repurposed drugs. The authors should compare the power of detecting true associations of GAM compared to standard LMM approaches.

**Response:** We agree it is important to compare the power of GAM and a standard GWAS LMM approach to identify mutations with true associations, particularly for the new and repurposed anti-TB drugs.

We thus compared the true positives (TP), false positives (FP), and false negatives (FN) mutations detected by GAM and GWAS LMM (**Figure 3e-g**), offering a pragmatic and robust assessment of the relative performance of these methods. This analysis determined that:

(1) For the new and repurposed drugs (BDQ, CFZ, DLM, LZD), GAM detected fewer true positives than the LMM, as expected given the small number of resistant isolates and the stratification required by GAM.
(2) For drugs with more drug-resistant isolates, however, the GAM and LMM methods often detected comparable numbers of true associations.
(3) Both approaches detected comparable numbers of false negative associations for each drug.
(4) However, GAM consistently detected substantially fewer false-positive associations (1-2 vs. 1000~5000) than the LMM approach across all analyzed drugs, due to its ability to filter cross-associations caused by overlapping drug resistances.

We included these data and highlighted these points in the revised manuscript (**Fig. 3e-g**) as below:

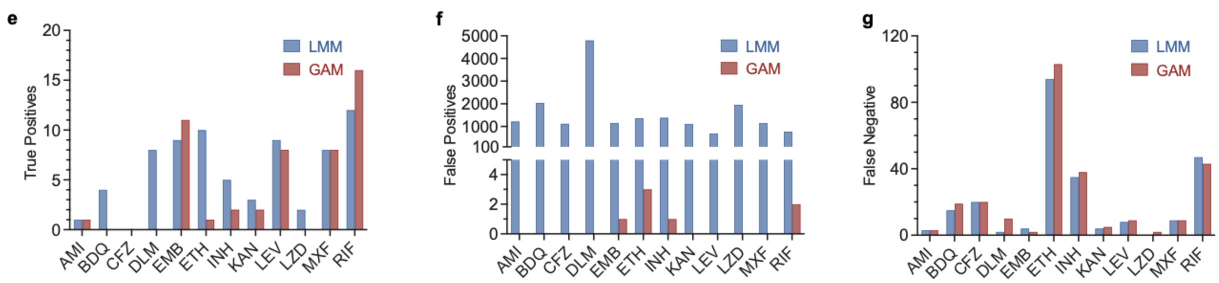



**Fig. 3: GAM and LMM detection of drug resistance associations. e** True positive, **f** false positive, **g** false negative numbers of mutations found by GWAS LMM (blue) and GAM (red).

(Page 8, Lines 170-180) "To evaluate the performance of GAM relative to standard GWAS LMM, true positives, false positives, and false negatives were compared between the two methods (**Fig. 3e-g**). Raw *Mtb* LMM results are available on Figshare (**SF3**). GAM consistently identified substantially fewer false positives than LMM (1-2 vs. 1000~5000) across all drugs in this comparison, highlighting its ability to avoid misleading cross-associations often caused by overlapping drug resistances. Both approaches detected comparable numbers of false negative associations for each drug (**Fig. 3g**). For new and repurposed drugs, GAM exhibited lower sensitivity (fewer true positives) than LMM, which was expected given the small number of resistant isolates and the stratification required by GAM. However, for drugs with larger sample sizes, GAM not only minimized false positives but also sometimes matched or outperformed LMM in detecting true associations, showcasing its precision when sufficient data is available. Overall, GAM showed more than a 200-fold higher positive prediction value (PPV) than LMM."

GAM's ability to effectively identify true associations while minimizing misleading ones is further validated through a performance comparison with LMM in the *S. aureus* dataset, using the same analytical framework applied to *Mtb*. The results, detailed in the revised manuscript (**Supplementary Table 4**), are comprehensively discussed in response to your major comment 2.

2) The concept of validating GAM or any other genotype-phenotype association method on antibiotic resistance datasets. The authors state that "To validate GAM performance, we next employed it to analyze a Staphylococcus aureus" dataset. However, pretty much most regression-based and ML approaches are expected to identify SNP resistance-associated and acquired AMR genes in such datasets, given the strong size effects of AMR genetic determinants in bacteria. As stated above, the authors should provide comparisons of methods in terms of power and false positive rates, and overall number of know AMR genetic determinants identified.

**Response:** We appreciate your concern that many regression-based and ML approaches can identify AMR-associated SNPs and genes in datasets with strong effect sizes.

We should have been more explicit in our statement because the intention of this analysis was not to claim that GAM outperforms LMM in the *S. aureus* dataset, but rather to demonstrate that GAM robustly detects resistance-associated SNPs and acquired AMR genes in datasets from other bacterial pathogens. To clarify this point, we have revised the text as follows:

(Page 8, Lines 181-182) "To examine GAM's ability to identify antimicrobial resistance (AMR) variants in other pathogens, we next employed it to analyze a *S. aureus* dataset…"

To provide the requested comparisons, we applied the same analysis framework used to compare GAM and LMM performance in the *Mtb* dataset to the *S. aureus* dataset and reported these results in the revised manuscript (**Supplementary Table 4**). Briefly, this data indicates that while both methods can effectively identify known *S. aureus* AMR determinants, GAM detects significantly fewer false positives, consistent with our findings in the *Mtb* dataset. Notably, this demonstrates that GAM can minimize misleading associations even in datasets with strong-effect AMR genes. We have added text to the manuscript to illustrate this point：

(Page 9, Lines 197-202 ) "GAM identified a comparable number of true positives while producing fewer false positives than LMM (**Supplementary Table 4**). This demonstrates that GAM outperformed LMM in detecting SNP-based drug resistance. However, for drugs such as penicillin, erythromycin, and clindamycin, where resistance primarily arises from horizontally transferred genes, LMM detected more true positives. Notably, this advantage came at the expense of a significantly higher number of false positives. All raw *S. aureus* LMM results are available on Figshare (**SF7** and **SF8**)."

**Supplementary Table 4. True positive (TP), false positive (FP), and false negative (FN) associations by GAM and LMM in *S. aureus*.**

| | TP | | FP | | FN | |
|---|---|---|---|---|---|---|
| | *GAM* | *LMM* | *GAM* | *LMM* | *GAM* | *LMM* |
| **GEN** | 1 | 1 | 16 | 5958 | 0 | 0 |
| **PEN** | 0 | 2 | 17 | 110017 | 3 | 0 |
| **MET** | 1 | 1 | 257 | 140196 | 0 | 0 |
| **FUS** | 10 | 10 | 16 | 87253 | 12 | 10 |
| **TEI** | 0 | 0 | 0 | 1360 | 0 | 0 |
| **VAN** | 0 | 0 | 0 | 579 | 0 | 0 |
| **ERY** | 0 | 2 | 74 | 94352 | 4 | 0 |
| **CLI** | 0 | 1 | 115 | 4356 | 4 | 0 |
| **LIN** | 0 | 0 | 0 | 866 | 1 | 1 |
| **CIP** | 6 | 4 | 700 | 150224 | 7 | 7 |
| **RIF** | 1 | 13 | 0 | 1905 | 15 | 1 |
| **TET** | 2 | 2 | 30 | 41393 | 1 | 0 |
| **TMP** | 5 | 3 | 47 | 8661 | 2 | 5 |

3) The authors should provide the raw GAM and LMM results per drug, that is, the table of all associated genetic variants and genes, ordered increasingly by p-value, not only the known AMR determinants presented in Supplementary Table 1 and 2. For example, it is unlikely that plasmid-borne AMR genes will be found associated on their own, but most likely along with other plasmid carried genes.

**Response:** Complete GAM and LMM results for *Mtb* and *S. aureus* AMR determinants per drug are now available. These results include all associated genetic variants and genes ranked by p-value and are provided in **Supplementary Table 1** and Figshare (**SF 6).**

We regret any lack of clarity on this point and should clarify how we presented the AMR determinants and the context of AMR in *Mtb* and *S. aureus*.

We have now updated **Supplementary Table 1** lists all the *Mtb* AMR determinants identified by GAM ordered by drug. Notably, *Mtb* does not typically harbor plasmids carrying drug-resistance genes. Unlike many bacterial species, drug resistance in *Mtb* arises primarily through chromosomal mutations under antibiotic selective pressure, rather than through horizontal gene transfer via plasmids [9, 18].

**Supplementary Table 2** shows the most relevant variants found by GAM along with what known HGT and SNP mechanisms are present in our dataset. We have now added a full list of *S. aureus* AMR determinants identified by GAM to Figshare (**S6)**.

We agree that the co-segregation of an AMR gene with other genes carried on the same plasmid could lead to the identification of false AMR associations for these additional passenger genes. However, the analyzed dataset does not include information on which genes are carried on the same plasmids, and such an analysis falls outside the scope of GAM.

[9] Cohen, K.A., Manson, A.L., Desjardins, C.A., Abeel, T. & Earl, A.M. Deciphering drug resistance in Mycobacterium tuberculosis using whole-genome sequencing: progress, promise, and challenges. Genome Med 11, 45 (2019).

[18] Dookie, N., Rambaran, S., Padayatchi, N., Mahomed, S. & Naidoo, K. Evolution of drug resistance in Mycobacterium tuberculosis: a review on the molecular determinants of resistance and implications for personalized care. *J Antimicrob Chemother* **73**, 1138-1151 (2018).

4) The section comparing GAM results to the WHO Mtb mutations catalogue (lines 174-221) is rather hard to follow. The authors show that training an ML model on all genetic variants from GAM detected genes outperforms the sensitivity of drug resistance predictions, when compared to the predictions made by GAM variants only, and the WHO 21 and 23 mutations catalogues (Figure 4d). However, the authors should also show the PPV and Specificity of the GAM+ML model in Figure 4b and 4c.

**Response:** We have updated **Fig. 4b-c** to include the PPV and specificity results for the GAM+ML model to provide a more comprehensive evaluation of its performance. We also include a table (**Supplementary Table 3**) that provides the quantitative results for the data as well as other relevant data metrics presented in **Fig. 4b-d**.

We have revised this section to improve its flow and clarity, highlighting GAM's ability to perform on par with the WHO mutation catalogue without requiring expert ruling or feature masking, and demonstrating how ML can enhance GAM's predictive performance. These changes are now reflected in the revised manuscript as below and we believe they significantly enhance the clarity and comprehensiveness of our analysis:

(Pages 10-14, Lines 212-258)

"**Optimization of variant detection to improve phenotypic prediction**

To evaluate predictive accuracy, DS2 dataset variants identified by GAM were compared to those of the WHO catalogue. GAM drug-specific variants demonstrated variable overlap with the WHO 2023 and 2021 *Mtb* mutation catalogues generated with the WHO dataset with (1.2% – 68.8%) and without (9.1% – 80.0%) interim criteria (**Supplementary Figs. 9-10**). Since GAM minimized the detection of misleading variants and requires no prior knowledge, it was hypothesized that GAM outputs could achieve better predictive accuracy and thus serve as improved inputs for a machine learning (ML) model (**Fig. 4a**). To assess the best model, nine ML models were screened and optimized for classification performance. Gradient Boosting achieved the highest mean accuracy (81.0%) and lowest overall variance (1.66%) across all nine drugs, (**Fig. 4b**), and this model was then introduced into a streamlined workflow to evaluate predictive performance (**Fig. 4c**).

PPV comparisons on AMR strains between the GAM, WHO, and GAM+ML approaches revealed differential predictive performance characteristics for these methods. GAM consistently minimized false-positive drug resistance strains by reducing cross-resistance associations. However, GAM PPV results varied by drug and differed from those of the WHO approach for amikacin, kanamycin, and rifampicin resistance (**Fig. 4d**). Integrating GAM results into a machine learning (ML) model enhanced the PPV for rifampicin resistance, without major effects on PPVs for other drugs resistances except for slight decreases in ethionamide, isoniazid, and levofloxacin.

Specificity analyses also emphasized the advantages of combining GAM with ML. GAM exhibited high specificity across all drugs, but the addition of ML led to a notable increase in rifampicin specificity (10.9%) (**Fig. 4e**). Specificity estimates modestly changed for other drugs, with minor decreases (ethionamide, isoniazid, and levofloxacin) and increases (ethambutol) observed for specific drugs. These findings indicate the robust predictive performance of GAM, and the enhancement provided by ML, particularly for rifampicin specificity. Notably, WHO+interim criteria did not markedly contribute to resistance classification since GAM and WHO non-interim variants produced similar PPV and specificity estimates (**Supplementary Fig. 11**).

Comparison of GAM, WHO, and GAM+ML sensitivity in predicting drug resistance revealed distinct performance patterns. GAM demonstrated variable sensitivity across different drugs, with the highest sensitivity observed for ethionamide resistance (**Fig. 4f**), while the WHO approach had better sensitivity for isoniazid resistance. Integrating GAM into an ML model improved predictive sensitivity for amikacin, ethionamide, isoniazid, kanamycin, and levofloxacin resistance, without affecting sensitivity estimates for other drugs. The use of ML thus enhances GAM predictive ability, particularly for drugs with complex resistance patterns. Area under the receiver operating characteristic curve (AUC) values ranged from 85.0 ± 1.0% to 97.0 ± 1.0% (**Supplementary Fig. 12**). Applying optimal cutoff values corresponding to the highest F1 scores increased overall model accuracy by 2.6%, further enhancing prediction of all nine drug phenotypes. These results demonstrate that integrating GAM outputs with ML can improve diagnostic accuracy, particularly for drugs where sensitivity and specificity remain challenging.

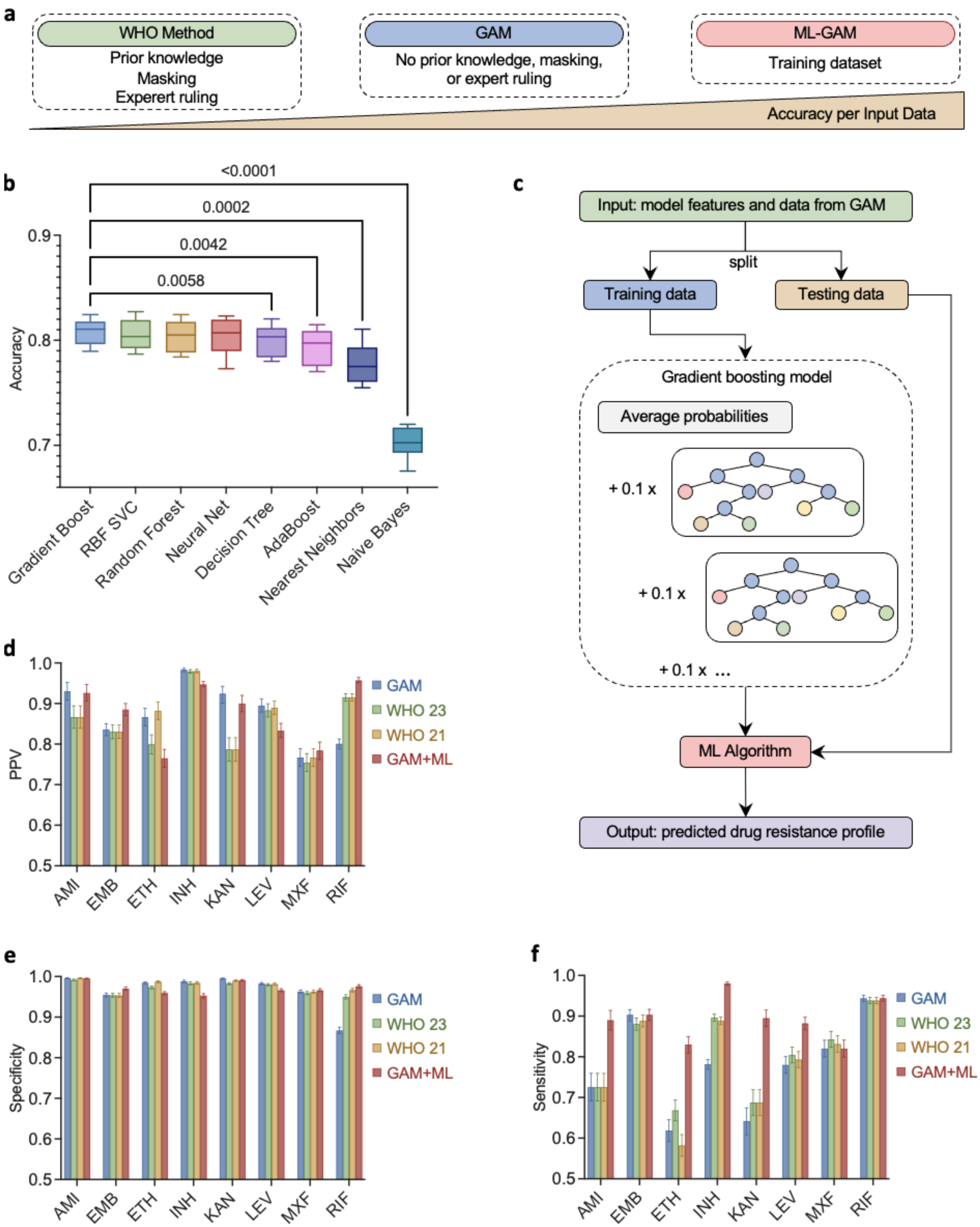



**Fig. 4: Gene variants associated with drug resistance and their predictive ability. a** Schematic comparing prior knowledge requirements and accuracy of different approaches. **b** GAM+ML classification accuracy for specific drug resistance using a ten-fold cross-validation approach. Data depict median (center bar), 25$^{th}$ and 75$^{th}$ percentile

(lower and upper box bounds), and minimum and maximum values (lower and upper whiskers), with p-values from repeat measure 1-way ANOVAs versus the Gradient Boosting reference model using Dunnett's test for multiple comparisons. **c** Workflow of the ML model using GAM variants as input. **d** PPV performance of predictive approaches applied to the CRyPTIC dataset, and **e** specificity and **f** sensitivity values for specific drug resistance using variants identified by GAM (blue); 2021 (yellow) and 2023 (green) WHO interim criteria; and a gradient boosting model using GAM variants (red). Error bars indicate 95% confidence intervals."

We have also revised and merged lines 174-180 (now lines 164-169) to improve the logical flow, positioning these sentences within the section that discusses the GAM workflow and its ability to recover known *Mtb* mutations from the WHO catalogue compared to LMM, followed by an analysis of GAM's performance relative to LMM.

5) The section on "Impact of sample size and data completeness on GAM performance" is very relevant and informative. As mentioned above, reporting the power and false positive rates with varying samples sizes of the GAM model, compared to commonly used LMM, will be informative too.

**Response:** We agree and appreciate your suggestion. We have updated **Fig. 5a** to show the true positive (TP) and false positive (FP) genes detected by the GAM and LMM models across varying sample sizes, providing a clearer comparison of their performance. Additionally, we have added **Fig. 5b** to present the positive predictive value (PPV) of genes detected at these sample sizes by each model. Furthermore, we have included supplementary data outlining the number of TP and FP SNPs detected across all drugs, excluding new and repurposed drugs (**Supplementary Fig. 14a**), and including them (**Supplementary Fig. 14c**). The supplementary data also includes a PPV ratio breakdown by sample size (**Supplementary Fig. 14b and 14d**) to better illustrate the models' performance across populations of different sizes. These results are discussed as follows:

(Page 14 lines 268-277) "Comparing GAM and LMM reveals that while LMM generally identified a slightly higher number of true positives, it exhibited a 100 to 1,000 times higher false positive rate than GAM across all sample sizes (**Fig. 5a**). Notably, PPV analysis further demonstrated that GAM consistently exhibited a significantly higher PPV than LMM (**Fig. 5b**), indicating its superior ability to distinguish true positives from false positives in the data. Similar trends were observed at the mutation level, where true positive differences remained minimal, but false positive rates and positive predictive value (PPV) declined more rapidly as sample size increased (**Supplementary Fig. 14a-b**). When new and repurposed drugs were included in the analysis, LMM showed an increase in true positives, while GAM struggled to capture significant associations due to the limited sample size available for these drugs; however, the LMM false positive rate also increased, while GAM's rate remained stable (**Supplementary Fig. 14c-d**)."

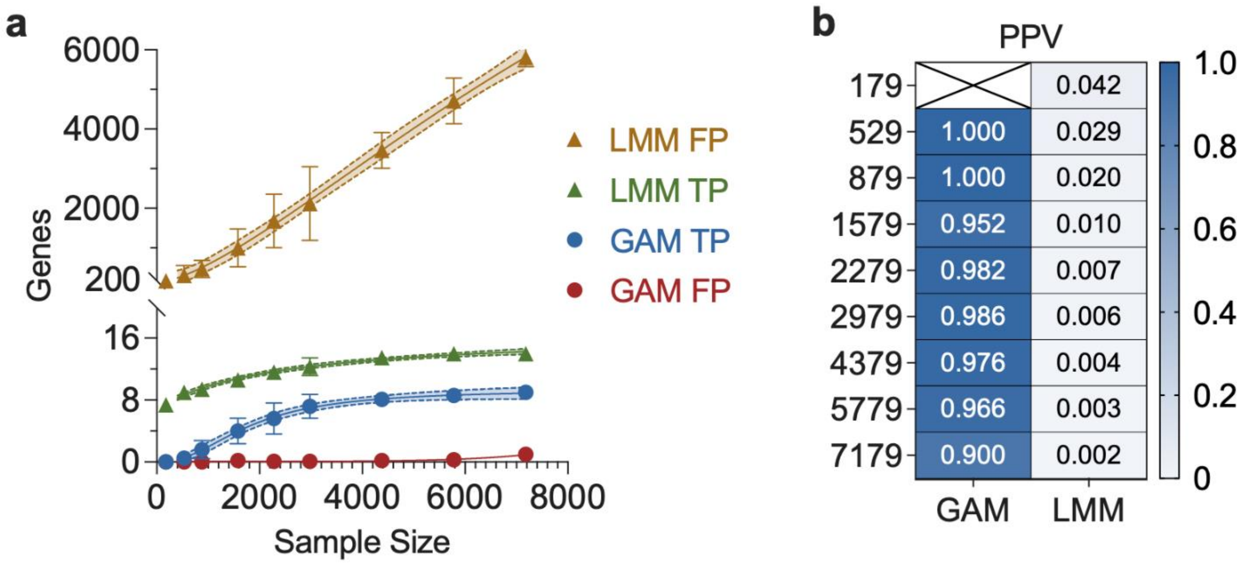
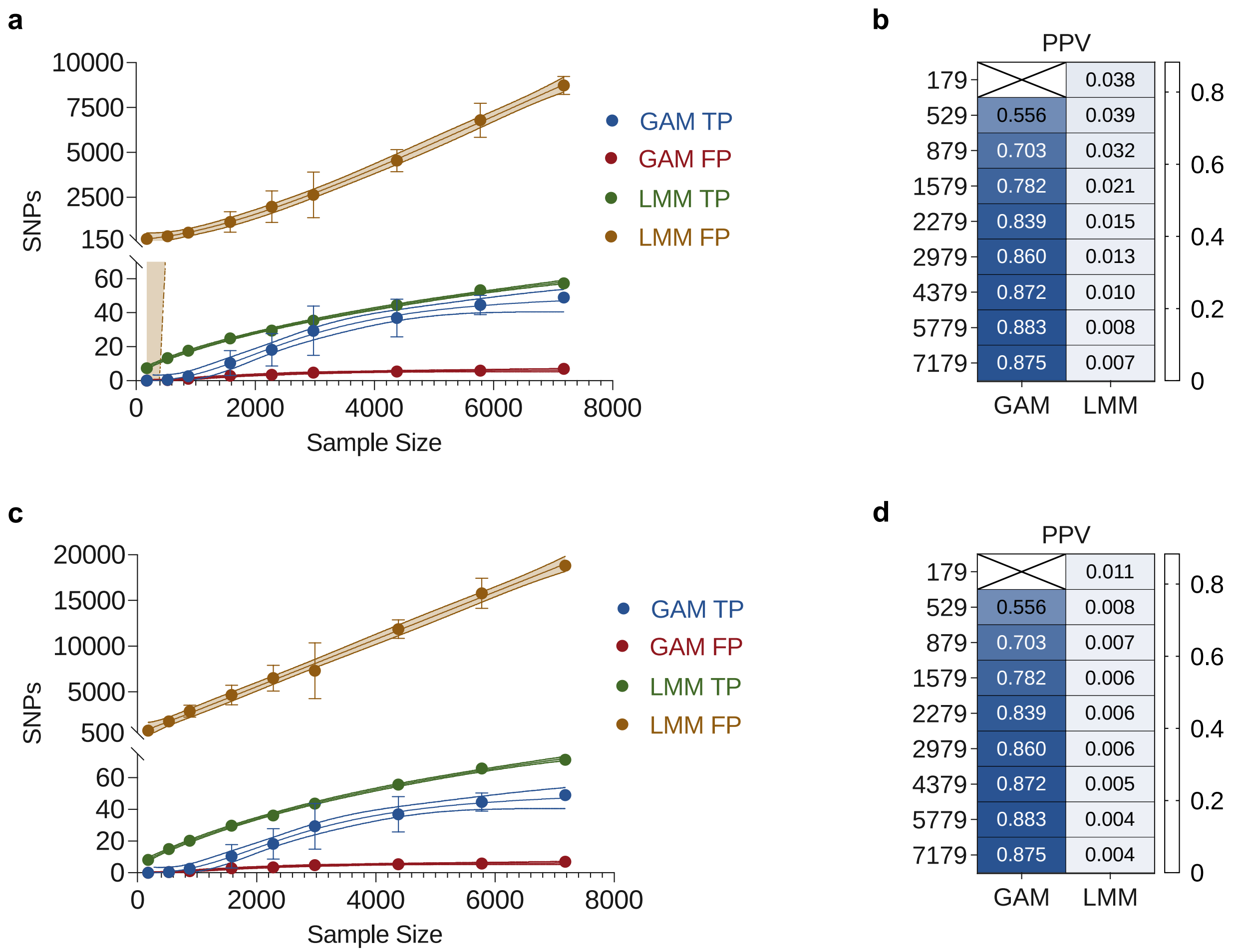



**Fig. 5: Effect of sample size and data incompleteness on GAM and ML-GAM outputs. a** Effect of sample size on GAM and LMM true positive (TP) and false positive (FP) gene identifications. Solid and dashed lines represent nonlinear sigmoidal curves and their 95% confidence intervals. Data points display mean ± standard error values for 10 replicates. Y-axis breakdown between 20 to 200. **b** Heatmap of mean GAM and LMM PPVs at the indicated sample sizes.




**Supplementary Fig. 14: Sample size effects on GAM and LMM identifications. a** Effect of sample size on GAM and LMM true positive (TP) and false positive (FP) SNP identifications without new and repurposed drugs, y-axis breakdown between 70 to 500, and **b** heatmap of mean GAM and LMM PPVs at the indicated sample sizes. **c** Effect of sample size on GAM and LMM true positive (TP) and false positive (FP) SNP identifications when including the new and repurposed drugs, y-axis breakdown between 75 to 500, and **d** heatmap of mean GAM and LMM PPVs at the indicated sample sizes. Solid and dashed lines represent nonlinear sigmoidal curves and their 95% confidence intervals. Data points display mean ± standard error values for 10 replicates.

6) The results presented in Figure 6 will be more informative if presented in terms of sensitivity and specificity rather than overall accuracy.

**Response:** We agree that presenting sensitivity and specificity results would provide more information. To address this, we retained **Fig. 6** as a concise summary statistic to avoid overloading the figure with 16 additional panels. Instead, we have added

**Supplementary Table 5**, which reports the true positives (TP), false positives (FP), false negatives (FN), true negatives (TN), sensitivity, and specificity obtained from machine learning models that used GAM, WHO 2023, and WHO 2021 variants as inputs for the dataset analyzed in **Fig. 6**.

**Supplementary Table 5. Predictive Performance of GAM+ML and WHO on clinical isolates.**

| Drug | Method | TP | FP | FN | TN | SEN | SPEC |
|---|---|---|---|---|---|---|---|
| **AMI** | *GAM* | 20 | 6 | 10 | 391 | 0.667 | 0.985 |
| | *WHO 23* | 20 | 6 | 10 | 391 | 0.667 | 0.985 |
| | *WHO 21* | 20 | 6 | 10 | 391 | 0.667 | 0.985 |
| **EMB** | *GAM* | 127 | 70 | 50 | 176 | 0.718 | 0.715 |
| | *WHO 23* | 123 | 70 | 54 | 176 | 0.695 | 0.715 |
| | *WHO 21* | 116 | 70 | 61 | 176 | 0.655 | 0.715 |
| **ETH** | *GAM* | 46 | 39 | 12 | 324 | 0.793 | 0.893 |
| | *WHO 23* | 46 | 41 | 12 | 322 | 0.793 | 0.887 |
| | *WHO 21* | 46 | 44 | 12 | 319 | 0.793 | 0.879 |
| **INH** | *GAM* | 206 | 5 | 62 | 79 | 0.769 | 0.940 |
| | *WHO 23* | 204 | 5 | 64 | 79 | 0.761 | 0.940 |
| | *WHO 21* | 203 | 5 | 65 | 79 | 0.757 | 0.940 |
| **KAN** | *GAM* | 30 | 11 | 8 | 378 | 0.789 | 0.972 |
| | *WHO 23* | 30 | 12 | 8 | 377 | 0.789 | 0.969 |
| | *WHO 21* | 30 | 14 | 8 | 375 | 0.789 | 0.964 |
| **LEV** | *GAM* | 6 | 5 | 6 | 95 | 0.500 | 0.950 |
| | *WHO 23* | 6 | 5 | 6 | 95 | 0.500 | 0.950 |
| | *WHO 21* | 6 | 5 | 6 | 95 | 0.500 | 0.950 |
| **MXF** | *GAM* | 51 | 41 | 28 | 295 | 0.646 | 0.878 |
| | *WHO 23* | 46 | 41 | 33 | 295 | 0.582 | 0.878 |
| | *WHO 21* | 41 | 40 | 38 | 296 | 0.519 | 0.881 |
| **RIF** | *GAM* | 68 | 3 | 22 | 92 | 0.756 | 0.968 |
| | *WHO 23* | 68 | 4 | 22 | 91 | 0.756 | 0.958 |
| | *WHO 21* | 68 | 5 | 22 | 90 | 0.756 | 0.947 |

7) Data and code availability: the GitHub repository github.com/jsaliba1/GAM does not exist, so the existence of code could not be reviewed. The section “Availability of data and materials” should also include the source of S. aureus genomes and linked AST data, as well as the genome accessions of the newly sequenced 428 DS3 isolates from China with linked DST profiles.

**Response:** We sincerely apologize for the citation error, which was inadvertently overlooked during our transition to uploading the code to CodeOcean, as recommended by Nature Communications to facilitate reviewer access. Unfortunately, we experienced repeated technical difficulties with the CodeOcean platform, which frequently and spontaneously deleted our code at various intervals after uploading, requiring multiple re-uploads. Due to these reliability issues, we have discontinued using CodeOcean and have reverted to GitHub. The GitHub repository (github.com/jsaliba1/GAM) is now open and contains the code along with all relevant data files for this manuscript.

We have also updated the "Availability of Data and Materials" (Line 606-613) section to include the source of the S. aureus genomes and their linked AST data, as well as the genome accessions of the newly sequenced isolates from China and their linked DST profiles.

**Minor comments**

1) Line 46. Spell out Mycobacterium tuberculosis (Mtb) in the first instance in the Introduction.

**Response:** We thank the reviewer for drawing our attention to this error, which we have corrected in the revised manuscript.

2) Lines 56-58. Drug resistance can arise on a background of pre-existing drug resistance. This is due to the nature of TB treatment being a combination therapy (multiple drugs used in regimens), and regimens being standardized for increasingly resistant strains (regimens for MDR and XDR strains). Please state and extend about how these treatment choices result in these overlapping resistances in clinical Mtb strains.

**Response:** We appreciate the opportunity to clarify the role of standardized TB treatment regimens in driving the development of overlapping AMR phenotypes.

We have added the following text to address this point:

(Page 2, lines 53 – 63) "Effective TB treatment requires the use of a multi-drug regimen that employs drugs with distinct mechanisms of action. Spontaneous mutations that confer resistance to one of drugs can allow *Mtb* bacilli to proliferate under selective pressure for mutations that confer resistance to other drugs in the multi-drug regimen.[18] Reliance of pre-defined drug regimens, including those tailored to resistant strains, can further promote development of multidrug-resistant (MDR-TB) or extensively drug-resistant (XDR-TB) *Mtb* strains.[19] This process drives sequential accumulation mutations in genes that directly confer resistance to specific drugs as well as in genes or sets of genes that alter metabolic processes to confer overlapping resistance to multiple drugs.[18,20] Mutations that produce overlapping resistance phenotypes can thus complicate efforts to genetically disentangle resistance mechanisms and identify specific mutations linked to resistance, particularly when a single mutation alters a pathway, or several pathways, that confers resistance to multiple drugs.[18] This complexity highlights the need for advanced analytical approaches that can address the dynamic and interconnected nature of the evolution of *Mtb* drug resistance."

[18] Dookie, N., Rambaran, S., Padayatchi, N., Mahomed, S. & Naidoo, K. Evolution of drug resistance in Mycobacterium tuberculosis: a review on the molecular determinants of resistance and implications for personalized care. J Antimicrob Chemother 73, 1138-1151 (2018).

[19] Rattan, A., Kalia, A. & Ahmad, N. Multidrug-resistant Mycobacterium tuberculosis: molecular perspectives. Emerg Infect Dis 4, 195-209 (1998).

[20] Waller, N.J.E., Cheung, C.Y., Cook, G.M. & McNeil, M.B. The evolution of antibiotic resistance is associated with collateral drug phenotypes in Mycobacterium tuberculosis. Nat Commun 14, 1517 (2023).

3) Line 271. Correct misspelt "ethanamide" drug name.

**Response:** We regret the error and have corrected it and others in the revised manuscript.

4) The GitHub repository github.com/jsaliba1/GAM does not exist, so the existence of code could not be reviewed.

**Response:** We sincerely apologize for this citation error, which was not corrected when we shifted to uploading this code to CodeOcean to provide limited access during the review process. However, CodeOcean proved unreliable, we have now made our code and all the associated data publicly accessible on GitHub at the following link:
github.com/jsaliba1/GAM

**RESPONSE TO Reviewer #2:**
The authors present an overall interesting approach of predicting multi-drug-resistance for Mtb samples from the CRyPTIC database without additional prior knowledge or expert rules. However, the detailed procedure seems unclear and intermediate results, specifically also in code / table format are missing for proper evaluation.

We thank the reviewer for their appreciation of our approach to identify mutations associated with *Mtb* isolates with multi-drug resistance, which does not rely on prior knowledge. We appreciate your feedback and the opportunity to clarify our methods and results.

**MAJOR COMMENTS**

1) Code and Method Clarity: Lack of access to the code and insufficient methodological details hinder a full evaluation, especially for GAM procedures and sample size/missing data handling.

**Response:** The GitHub repository (github.com/jsaliba1/GAM) is now open and contains the code along with all relevant data files for this manuscript.

We also apologize for the lack of clarity in our methods. We included a detailed description of GAM procedure and group criteria (further detailed in response to your minor comments 2 and 5), the result interpretation from mutation level to gene level (further detailed in response to minor comment 5), performance evaluation and comparison (detailed in response to minor comment 6), as well as the missing size/missing data handling (detailed in response minor comment 8).

2) GAM Output and Validation: The selection criteria for resistance groups, exact GAM output, and its performance evaluation (particularly against the WHO catalog) need clarification, ideally with validation on independent or held-out data.

**Response:** Resistance groups analyzed by GAM were defined based on drug susceptibility testing (DST) profiles, where isolates sharing identical resistance patterns were grouped together. To ensure statistical robustness, we only considered resistance groups that included two or more *Mtb* strains to maximize both the number and diversity of resistance groups. We have clarified the selection process in the revised manuscript (for details, see response to your minor comment 2).

The GAM output is a dataframe which lists genetic variants that are associated with a specific single drug resistance phenotype and the p-values for these associations. (for more details, see minor comment 6).

We use several methods to evaluate GAM's performance. This included the use of held-out data (**Supplementary Figures 17 and 18**), and a validation study performed on an independent multi-site clinical dataset (**Figure 6 and Supplementary Figure 19**). (A more detailed discussion about these data can be seen in the response to your minor comment 6)

3) Supplemental Data and Consistency: Complete tables of quantitative results, especially significant variants for GAM groups, should be provided as supplemental data. Consistent terminology for drug names (e.g., Rifampin vs. Rifampicin) is also needed.

**Response:** We have added a list of all significant variants detected in each GAM *Mtb* groups to Figshare (**SF2**). Similarly, lists of significant variants detected in *S. aureus* groups have been added to Figshare (**SF4** and **SF5**). We have also added **Supplementary Table 3** that supplements the results presented in **Figure 4b-d** by listing the true and false positives and true and false negatives identified in these analyses; **Supplementary Table 4** comparing the performance of GAM and an LMM model when analyzing an *S. aureus* dataset; **Supplementary Table 5** comparing the sensitivity and specificity obtained from machine learning models using GAM versus WHO inputs for each analyzed drug (**Figures 6b-i**). Finally, a list of variants identified by LMM, along with their p-values for specific single drug associations for *Mtb*, has been added to Figshare (**SF3**), and similar lists for *S. aureus* have been added to Figshare (**SF7** and **SF8**). These additions are further detailed in our response to your minor comments 3 and 4.

Finally, we have now revised the manuscript to replace all mentions of "Rifampin" with "Rifampicin".

4) GAM Generalization and Horizontal Gene Transfer: The manuscript lacks clarity on GAM's generalization for S. aureus and the limitations in detecting horizontal gene transfer, requiring further explanation.

**Response:** We appreciate the opportunity to clarify GAM's generalizability for *S. aureus* and its limitations for detecting horizontal gene transfer (HGT) associated with antimicrobial resistance (AMR). GAM was designed to detect genomic variants associated with AMR in *Mtb* genomic sequence datasets, and we acknowledge that its application to *S. aureus* requires careful consideration, while the overall framework of GAM remains the same, there are inherent differences in the genetics and resistance mechanisms of these two species. Specifically, *S. aureus* has greater genetic diversity and exhibits frequent HGT, which GAM is not specifically designed to detect (see details in the response to your minor comment 10).

We now clarify that GAM operates by identifying SNPs that are statistically associated with drug resistance phenotypes, and thus does not directly detect HGT events, which involve the transfer of entire genes or genomic regions between strains, unless these transfer events are associated with an SNP linked to an AMR phenotype.

We address this limitation in the third paragraph of the Discussion (page 20, lines 391-406), where we indicate that GAM's association-based approach cannot effectively capture HGT events since it relies on a SNP-level analysis and the use of a single reference genome. We hypothesize that GAM might detect HGT in two scenarios by: (1) identifying SNPs within horizontally transferred genes that are linked to resistance, and (2) detecting polymorphisms near horizontally transferred genes if these polymorphisms are associated with resistance.

For more details, see minor comments below.

**MINOR COMMENTS**

1) The code repository was not available to the reviewer, therefore it couldn't be used to evaluate procedures with unclear description in the methods.

**Response:** We apologize that this repository was not available for review. We previously submitted our code to CodeOcean per the recommendation of Nature Communications. We experienced repeated technical difficulties with the CodeOcean platform, which frequently and spontaneously deleted our code at various intervals after uploading, requiring multiple re-uploads. Due to these reliability issues, we have discontinued using CodeOcean and have reverted to GitHub. The GitHub repository (github.com/jsaliba1/GAM) is now open and contains the code along with all relevant data files for this manuscript.

2) How were the different resistance groups evaluated in the GAM selected? All possible combinations that were observed?

**Response:** Sorry for any confusion, the following text has been added to the methods section of the revised manuscript to clarify this point:

(Page 22, lines 455-462) “CRyPTIC database entries that did not meet study exclusion criteria (**Supplementary Fig. 1**) were grouped based on their unique drug resistance profiles to produce a control group that contained the genetic variations present in isolates susceptible to all 13 drugs, and an array of groups with distinct drug resistance profiles and associated genetic variants. Resistance groups analyzed by GAM were defined based on drug susceptibility testing (DST) profiles, where isolates sharing identical drug resistance profiles were grouped together. To ensure statistical robustness, we selected all observed resistance groups that included two or more *Mtb* strains to maximize both the number and diversity of resistance groups. Non-synonymous genetic variants that differed from the *Mtb* H37Rv reference genome were identified in all groups.”

3) Association results of all variants, but specifically significant variants for all GAM groups should be supplied as Supplemental Data.

**Response:** Thank you for your feedback. We have now added a list of all variants detected in each GAM *Mtb* groups to Figshare (**SF2**). Additionally, we have added lists of all SNPs and genes detected in each GAM *S. aureus* group to Figshare (**SF4** and **SF5**, respectively).

4) Similar tables of complete quantitative results should be supplied for all following analyses as well, especially also for data shown in the figures.

**Response:** We have added the following supplementary tables to supply this information:

(1) **Supplementary Table 3** provides the quantitative results for the data presented in **Figures 4b-d**, as well as other relevant data metrics.

**Supplementary Table 3. Performance of GAM, GAM+ML, and WHO in predicting drug resistance in *Mtb*.**

| Drug | Method | TP | FP | FN | TN | PPV | PPV_uci | PPV_lci | SEN | SEN_uci | SEN_lci | SPEC | SPEC_uci | SPEC_lci |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AMI | *GAM* | 481 | 36 | 182 | 9328 | 0.930 | 0.952 | 0.908 | 0.725 | 0.759 | 0.692 | 0.996 | 0.997 | 0.995 |
| | *GAM+ML* | 590 | 47 | 73 | 9317 | 0.926 | 0.853 | 0.820 | 0.890 | 0.917 | 0.889 | 0.995 | 0.960 | 0.950 |
| | *WHO 23* | 481 | 77 | 182 | 9287 | 0.862 | 0.887 | 0.843 | 0.725 | 0.645 | 0.593 | 0.992 | 0.987 | 0.982 |
| | *WHO 21* | 481 | 36 | 182 | 9328 | 0.930 | 0.987 | 0.979 | 0.725 | 0.793 | 0.770 | 0.996 | 0.991 | 0.986 |
| EMB | *GAM* | 1639 | 321 | 176 | 6775 | 0.836 | 0.946 | 0.903 | 0.903 | 0.674 | 0.609 | 0.955 | 0.997 | 0.994 |
| | *GAM+ML* | 1639 | 213 | 176 | 6883 | 0.885 | 0.911 | 0.879 | 0.903 | 0.800 | 0.760 | 0.970 | 0.985 | 0.980 |
| | *WHO 23* | 1598 | 327 | 217 | 6769 | 0.830 | 0.789 | 0.745 | 0.880 | 0.841 | 0.799 | 0.954 | 0.967 | 0.959 |
| | *WHO 21* | 1612 | 329 | 203 | 6767 | 0.830 | 0.814 | 0.790 | 0.888 | 0.951 | 0.936 | 0.954 | 0.876 | 0.859 |
| ETH | *GAM* | 804 | 126 | 495 | 7931 | 0.865 | 0.947 | 0.906 | 0.619 | 0.914 | 0.866 | 0.984 | 0.996 | 0.994 |
| | *GAM+ML* | 1078 | 331 | 221 | 7726 | 0.765 | 0.900 | 0.870 | 0.830 | 0.917 | 0.889 | 0.959 | 0.974 | 0.966 |
| | *WHO 23* | 868 | 210 | 431 | 7847 | 0.805 | 0.787 | 0.743 | 0.668 | 0.850 | 0.809 | 0.974 | 0.963 | 0.955 |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *WHO 21* | 756 | 101 | 543 | 7956 | 0.882 | 0.954 | 0.941 | 0.582 | 0.984 | 0.976 | 0.987 | 0.958 | 0.947 |
| **INH** | *GAM* | 3655 | 62 | 1022 | 5286 | 0.983 | 0.920 | 0.880 | 0.781 | 0.915 | 0.874 | 0.988 | 0.993 | 0.989 |
| | *GAM+ML* | 4583 | 254 | 94 | 5094 | 0.947 | 0.851 | 0.816 | 0.980 | 0.897 | 0.866 | 0.953 | 0.970 | 0.963 |
| | *WHO 23* | 4192 | 89 | 485 | 5259 | 0.979 | 0.805 | 0.762 | 0.896 | 0.841 | 0.799 | 0.983 | 0.970 | 0.963 |
| | *WHO 21* | 4155 | 84 | 522 | 5264 | 0.980 | 0.964 | 0.951 | 0.888 | 0.951 | 0.937 | 0.984 | 0.980 | 0.972 |
| **KAN** | *GAM* | 536 | 44 | 299 | 9206 | 0.924 | 0.847 | 0.813 | 0.642 | 0.895 | 0.866 | 0.995 | 0.959 | 0.949 |
| | *GAM+ML* | 747 | 83 | 88 | 9167 | 0.900 | 0.847 | 0.813 | 0.895 | 0.895 | 0.866 | 0.991 | 0.959 | 0.949 |
| | *WHO 23* | 574 | 162 | 261 | 9088 | 0.780 | 0.829 | 0.782 | 0.687 | 0.694 | 0.643 | 0.982 | 0.977 | 0.970 |
| | *WHO 21* | 574 | 93 | 261 | 9157 | 0.861 | 0.983 | 0.975 | 0.687 | 0.905 | 0.888 | 0.990 | 0.987 | 0.980 |
| **LEV** | *GAM* | 1260 | 148 | 355 | 8351 | 0.895 | 0.899 | 0.867 | 0.780 | 0.824 | 0.785 | 0.983 | 0.983 | 0.977 |
| | *GAM+ML* | 1424 | 285 | 191 | 8214 | 0.833 | 0.899 | 0.867 | 0.882 | 0.824 | 0.785 | 0.966 | 0.983 | 0.977 |
| | *WHO 23* | 1299 | 172 | 316 | 8327 | 0.883 | 0.776 | 0.732 | 0.804 | 0.863 | 0.823 | 0.980 | 0.963 | 0.955 |
| | *WHO 21* | 1281 | 159 | 334 | 8340 | 0.890 | 0.923 | 0.905 | 0.793 | 0.946 | 0.931 | 0.981 | 0.955 | 0.944 |
| **MXF** | *GAM* | 1076 | 327 | 236 | 8500 | 0.767 | 0.952 | 0.908 | 0.820 | 0.759 | 0.692 | 0.963 | 0.997 | 0.995 |
| | *GAM+ML* | 1076 | 297 | 236 | 8530 | 0.784 | 0.847 | 0.814 | 0.820 | 0.903 | 0.874 | 0.966 | 0.959 | 0.949 |
| | *WHO 23* | 1106 | 360 | 206 | 8467 | 0.754 | 0.904 | 0.861 | 0.843 | 0.609 | 0.555 | 0.959 | 0.990 | 0.985 |
| | *WHO 21* | 1091 | 332 | 221 | 8495 | 0.767 | 0.984 | 0.976 | 0.832 | 0.897 | 0.879 | 0.962 | 0.988 | 0.981 |
| **RIF** | *GAM* | 3438 | 851 | 205 | 5558 | 0.802 | 0.887 | 0.834 | 0.944 | 0.719 | 0.656 | 0.867 | 0.992 | 0.988 |
| | *GAM+ML* | 3439 | 154 | 204 | 6255 | 0.957 | 0.906 | 0.873 | 0.944 | 0.813 | 0.773 | 0.976 | 0.984 | 0.978 |
| | *WHO 23* | 3419 | 322 | 224 | 6087 | 0.914 | 0.789 | 0.745 | 0.939 | 0.852 | 0.811 | 0.950 | 0.966 | 0.958 |
| | *WHO 21* | 3419 | 213 | 224 | 6196 | 0.941 | 0.949 | 0.934 | 0.939 | 0.946 | 0.931 | 0.967 | 0.971 | 0.962 |

(2) **Supplementary Table 4** compares the results of the *S. aureus* GAM and LMM models.

**Supplementary Table 4. True positive (TP), false positive (FP), and false negative (FN) associations by GAM and LMM in *S. aureus*.**

| | TP | | FP | | FN | |
|---|---|---|---|---|---|---|
| | *GAM* | *LMM* | *GAM* | *LMM* | *GAM* | *LMM* |
| **GEN** | 1 | 1 | 16 | 5958 | 0 | 0 |
| **PEN** | 0 | 2 | 17 | 110017 | 3 | 0 |
| **MET** | 1 | 1 | 257 | 140196 | 0 | 0 |
| **FUS** | 10 | 10 | 16 | 87253 | 12 | 10 |
| **TEI** | 0 | 0 | 0 | 1360 | 0 | 0 |
| **VAN** | 0 | 0 | 0 | 579 | 0 | 0 |
| **Ery** | 0 | 2 | 74 | 94352 | 4 | 0 |
| **CLI** | 0 | 1 | 115 | 4356 | 4 | 0 |
| **LIN** | 0 | 0 | 0 | 866 | 1 | 1 |
| **CIP** | 6 | 4 | 700 | 150224 | 7 | 7 |
| **RIF** | 1 | 13 | 0 | 1905 | 15 | 1 |
| **TET** | 2 | 2 | 30 | 41393 | 1 | 0 |
| **TMP** | 5 | 3 | 47 | 8661 | 2 | 5 |

(3) **Supplementary Table 5**, which compares results obtained by machine learning models employing GAM or WHO inputs (complimentary to results presented in **Figures 6b-i**).

**Supplementary Table 5. GAM+ML and WHO model predictive performance using data from clinical isolates from a multi-site *Mtb* cohort enrolled in China.**

| Drug | Method | TP | FP | FN | TN | SEN | SPEC |
|---|---|---|---|---|---|---|---|
| AMI | *GAM* | 20 | 6 | 10 | 391 | 0.667 | 0.985 |
| | *WHO 23* | 20 | 6 | 10 | 391 | 0.667 | 0.985 |
| | *WHO 21* | 20 | 6 | 10 | 391 | 0.667 | 0.985 |
| EMB | *GAM* | 127 | 70 | 50 | 176 | 0.718 | 0.715 |
| | *WHO 23* | 123 | 70 | 54 | 176 | 0.695 | 0.715 |
| | *WHO 21* | 116 | 70 | 61 | 176 | 0.655 | 0.715 |
| ETH | *GAM* | 46 | 39 | 12 | 324 | 0.793 | 0.893 |
| | *WHO 23* | 46 | 41 | 12 | 322 | 0.793 | 0.887 |
| | *WHO 21* | 46 | 44 | 12 | 319 | 0.793 | 0.879 |
| INH | *GAM* | 206 | 5 | 62 | 79 | 0.769 | 0.940 |
| | *WHO 23* | 204 | 5 | 64 | 79 | 0.761 | 0.940 |
| | *WHO 21* | 203 | 5 | 65 | 79 | 0.757 | 0.940 |
| KAN | *GAM* | 30 | 11 | 8 | 378 | 0.789 | 0.972 |
| | *WHO 23* | 30 | 12 | 8 | 377 | 0.789 | 0.969 |
| | *WHO 21* | 30 | 14 | 8 | 375 | 0.789 | 0.964 |
| LEV | *GAM* | 6 | 5 | 6 | 95 | 0.500 | 0.950 |
| | *WHO 23* | 6 | 5 | 6 | 95 | 0.500 | 0.950 |
| | *WHO 21* | 6 | 5 | 6 | 95 | 0.500 | 0.950 |
| MXF | *GAM* | 51 | 41 | 28 | 295 | 0.646 | 0.878 |
| | *WHO 23* | 46 | 41 | 33 | 295 | 0.582 | 0.878 |
| | *WHO 21* | 41 | 40 | 38 | 296 | 0.519 | 0.881 |
| RIF | *GAM* | 68 | 3 | 22 | 92 | 0.756 | 0.968 |
| | *WHO 23* | 68 | 4 | 22 | 91 | 0.756 | 0.958 |
| | *WHO 21* | 68 | 5 | 22 | 90 | 0.756 | 0.947 |

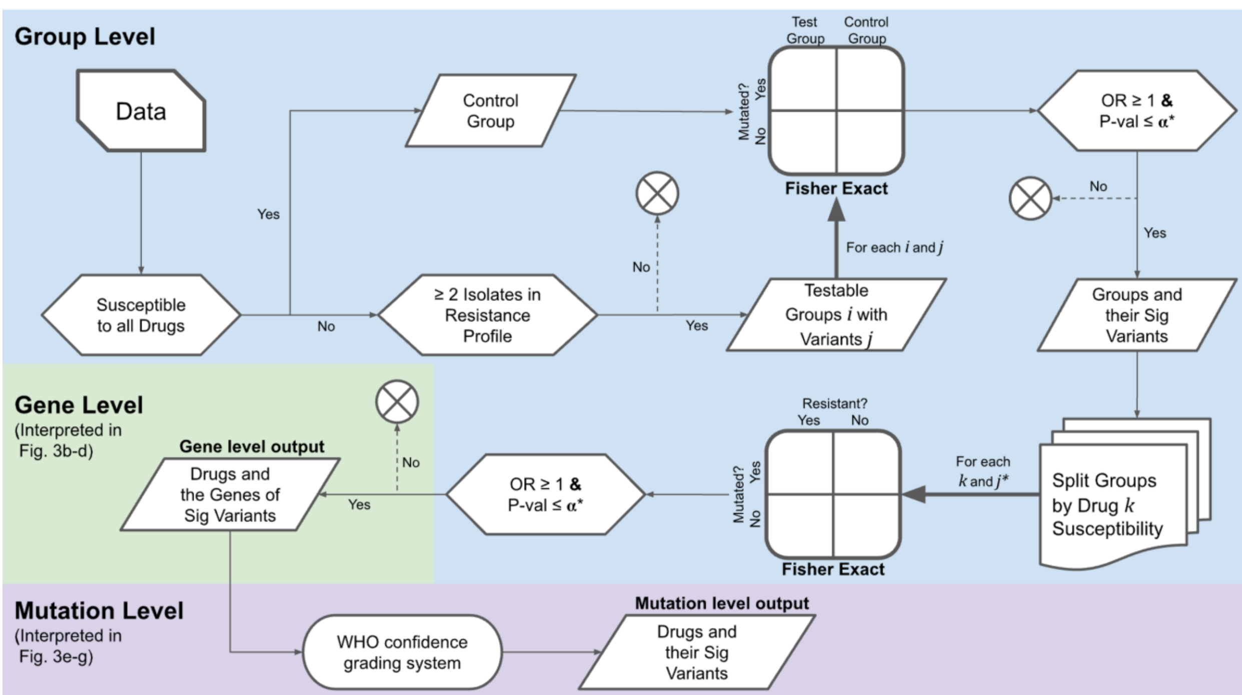
(4) Figshare files that lists the raw *Mtb* LMM results per drug (**SF3**), the raw *S. aureus* LMM SNP results per drug (**SF7**), and the raw *S. aureus* LMM gene results per drug (**SF8**).

5) How are gene level associations evaluated in the GAM procedure? From Figure 3 A, the Supplemental Figure 2 and the methods (line 409-411), this is unclear. It seems like the outline is switching from variant level to gene level and back to variant level.

**Response:** We apologize for the lack of clarity on this key point and have now combined and reordered the text presented in the method sections entitled "**GAM analysis procedure"** and **"Identification of AMR-associated sequence variants**" to make sure they align with the workflow presented in **Fig. 3a**. Additionally, we have added text to this combined section to further clarify the GAM procedure, the text now reads:

(Page 23, lines 470-474) "The GAM procedure identifies DNA variants that are significantly associated with specific single drug AMR phenotypes, while results are interpreted at both the gene and sequence variants level. AMR-associated variants are first linked with their respective genes and then all mutations within these genes are analyzed to identify those that are significantly associated with the targeted single-drug AMR phenotype."

We have also revised **Supplementary Figure 2** and its figure legend:




**"Supplementary Fig. 2: Flow chart of the GAM procedure.** Flow chart of conditional statements (hexagons) and internal data repositories (parallelograms). **Group level:** All groups containing ≥2 isolates are generated according to their unique resistance profiles. Isolates susceptible to all drugs are added to the control group, and those with unique drug-resistance profiles are excluded from analysis. All drug-resistant groups i and their

DNA variants j are then compared to the control group by Fisher's exact tests corrected for multiple tests, retaining only those variants that are significantly enriched in the drug-resistant groups. **Gene level interpretation:** All significant DNA variants j* belonging to groups resistant to a given drug k are then analyzed using Fisher's exact tests to identify significant drug-SNP interactions after correcting for multiple tests were used to identify drug-gene interactions are then linked with their respective genes. **Mutation level output and interpretation:** All variants mapping to these gene associations are then analyzed by a WHO confidence grading system [3] to identify rarer variants associated with drug resistance."

[3] WHO. Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance. (2021).

6) What was the exact output of the GAM procedure and how was it evaluated for performance? Was the WHO catalog used as ground truth/gold standard?
Predictive performance should be evaluated on an independent dataset. At the least, performance needs to be evaluated using unseen, held out data.

**Response:** The GAM output is a data frame which lists genetic variants that are associated with a specific single drug resistance phenotype and the p-values for these associations. GAM results were compared to AMR mutations listed in the WHO *Mtb* mutation catalogue, which served as a reference for genes associated with *Mtb* drug resistance and is a broadly accepted catalogue of resistance-associated mutations.

We used several methods to evaluate GAM's performance, we have now added a section to the methods to clarify this point:

(Page 24, Lines 503-509) "To evaluate the accuracy of GAM identification of variants associated with resistance, GAM and GWAS LMM outputs were compared for true positive, false positive, and true negative counts based on known variant-resistance associations summarized in the WHO Mtb mutation catalogue, which is the clinically recognized gold standard reference for resistance-associated *Mtb* mutations. [35,36]

Additionally, we assessed predictive performance by comparing the sensitivity, specificity, and PPV of GAM outputs versus the WHO mutation catalogue as predictors of phenotypic drug susceptibility across different datasets."

We apologize for any confusion and would like to clarify that we have evaluated the predictive performance of the GAM+ML model using held-out data from DS3 isolates that could not be used for grouping in GAM analysis (**Supplementary Figures 17 and 18**). We have also made a substantial effort to curate an independent muti-site clinical dataset

(DS4) to evaluate predictive performance (**Figure 6 and Supplementary Figure 19**). We agree that it would be desirable to perform a validation study on another large independent *Mtb* dataset, but most large publicly available *Mtb* datasets, such as the WHO *Mtb* dataset, contain data that is also present in the CRyPTIC dataset or lack DST phenotyping data required for this analysis.

[35] WHO. Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance.  (2021).

[36] WHO. Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance second edition. (2023).

7) The description of the generalization of the GAM for S. aureus is insufficient and unclear. Specifically, what was the “global assessment” as mentioned in comparisons between groups and controls were substituted with a global assessment of whether mutations differed from the control group.” (line 432-433)?

**Response:** We regret the lack of clarity on this point. GAM analyses normally compare SNPs in the control group against those in the resistance groups to identify SNPs that are detected at significantly higher frequency in the resistance groups and then compare all the SNPs detected in the groups to resistant to a specific drug against all those in the groups sensitive to that drug to identify SNPs associated with resistance to a specific drug. However, our *S. aureus* dataset lacked a large, well-defined control group, and thus did not have sufficient statistical power to detect significant differences between the control and resistance groups after correction for multiple testing. We therefore instead performed a "global assessment," where we first compared the resistance groups against each other to identify SNPs that revealed significant frequency differences between resistance groups that were sensitive or resistant to a specific drug, and these compared these SNPs to groups to the control group. By reversing the order of comparisons, we reduced the number of tests performed between the control and resistance groups, which allowed us to identify significant differences. We have revised the manuscript to clarify this approach:

(Page 24. Line 294-497) “However, given the smaller size of the overall sample and its control group, the order of GAM analyses was reversed, so that resistance groups, were first compared to identify SNPs that were detected at different frequencies in the resistance groups that were resistant and sensitive to a specific drug, and these SNPs were then compared to the control group to exclude neutral mutations.”

8) Since basic methods are unclear, it is difficult to assess if the sample size and missing data investigations are reasonable.

**Response:** We apologize for any confusion and have revised the methods to clarify our approach. Briefly, to assess the effect of sample size on GAM performance, we randomly selected isolates from the overall CRyPTIC dataset to generate datasets containing 179 to 7197 isolates and then performed GAM analyses to identify true- and false-positive drug-gene associations for each sample size, analyzing ten independent replicates for each sample size. To assess the effect of missing data on GAM performance, we randomly excluded drug resistance entries from the dataset, deleting from 1% to 25% of these entries among the isolates in the sample, so that individual isolates could have variable amounts of missing data. We repeated this process to generate ten independent samples for each missing data condition and determined the mean and standard deviation of the number of accurate gene identifications for each missing data rate.

To address concerns about the sample size and missing data investigations, the revised manuscript now includes an expanded description of the GAM procedure, its outputs, and performance evaluation (as detailed in our responses to minor comments 5-6). Additionally, we have provided more detail on the investigations related to the impact of sample size and data completeness (Page 26, Lines 557-577), and these updates are highlighted in yellow in the revised manuscript.

In addition, we have incorporated a comparative analysis of GAM and LMM performance across datasets of varying sizes (**Fig. 5a and 5b**). This analysis used a FaST-LMM model to evaluate the same data analyzed by GAM, emphasizing the effects of sample size. These revisions provide a clearer rationale for the study's focus on sample size and data completeness and underscore GAM's percision in consistently identifying true associations while minimizing false positives, regardless of dataset size.

9) Throughout the manuscript, Rifampin and Rifampicin are used. The authors should keep using only one name for one drug.

**Response:** We apologize for this inconsistency and have now revised the manuscript to replace all mentions of “Rifampin” with "Rifampicin".

10) Would it be possible to clarify, why it should be challenging to detect horizontal gene transfer with the GAM, as it is based on associations and not e.g. phylogenetic trees?

**Response:** We appreciate the opportunity to clarify GAM's generalizability for *S. aureus* and its limitations for detecting HGT associated with AMR. GAM was designed to detect genomic variants associated with AMR in *Mtb* genomic sequence datasets, and we acknowledge that its application to *S. aureus* requires careful consideration, while the overall framework of GAM remains the same, there are inherent differences in the genetics and resistance mechanisms of these two species. Specifically, *S. aureus* has greater genetic diversity and exhibits frequent HGT, which GAM is not specifically designed to detect. However, HGT typically involves the transfer of entire genes or larger genomic regions between isolates. These transferred elements are often extra-genomic and thus may not be represented in the reference genome used by GAM. If the transferred genes do not contain SNPs detectable in the genomic sequences analyzed, they will not be detected by GAM.

The reference genome we used in our analysis does not include data from horizontally transferred genes. We supplemented the data used by GAM with a list of genes present in each isolate, but this does not allow GAM to directly detect HGT events as there is no SNP data available from these genes, which GAM relies on to identify drug-gene associations. We have added text to the Methods to clarify this point:

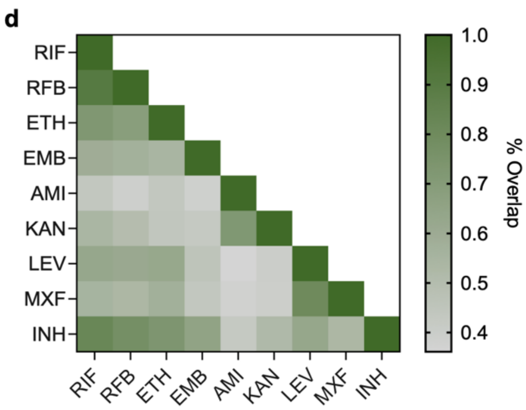
(Page 24. Lines 499-501) "For instances where drug resistance mechanisms stemmed from gene acquisition rather than mutation, a dataset of all genes present within each isolate was used and the analysis terminated at the gene level, as further data extraction was not feasible."

Moreover, GAM's association-based detection methodology does not evaluate evolutionary context that can be important for detecting HGT. For example, phylogenetic methods can identify incongruences in gene tree topologies that suggest HGT, but GAM exclusively uses SNP-level correlations to predict phenotypic outcomes. GAM is therefore not designed to effectively detect the horizontal transfer of resistance genes.

We address this limitation in the third paragraph of the Discussion (page 20, lines 391-406), where we indicate that GAM's association-based approach cannot effectively capture HGT events since it relies on a SNP-level analysis and the use of a single reference genome. We hypothesize that GAM might detect HGT in two scenarios by: (1) identifying SNPs within horizontally transferred genes that are linked to resistance, and (2) detecting polymorphisms near horizontally transferred genes if these polymorphisms are associated with resistance.

11) Figure 3d: legend for the color is missing.

**Response:** We apologize for the oversight. We have added the missing colour scale legend, which indicates the percent overlap between drug resistance phenotypes, with darker shades indicating stronger overlap and lighter shades indicating weaker or no overlap. The text now reads:




"**Fig. 3d** Co-occurrence of DS2 drug-resistant phenotypes, where dark and light green indicates high and low percent overlap, respectively."

12) WHO catalog contents and expert ruling should be explained very briefly in the context on how that approach differs from the GAM/ML analyses.

**Response:** We have now added a brief description of the WHO catalogue data format and expert rules, and how the process used to identify the WHO variants differs from the means through which AMR variants are identified in the GAM/ML analyses. Briefly, the WHO catalogue uses an algorithm with multiple criteria to identify AMR-associated mutations. These include the use of solo mutations (single mutations in a gene or its promoter) present in more than five isolates, and selection based on a lower bound of a mutation 95% CI PPV interval (≥ 25%), odds ratio (<1), and statistical significance ($p \leq 0,05$) by Fisher exact FDR test, although additional mutations that met more relaxed conditions (present in two or more resistant isolates with a PPV ≥ 50%). Since this process relies on the use of solo mutations, variants present in a gene with multiple mutations cannot analyzed for their potential contribution to AMR, significantly limiting the amount of data available for analysis. Expert rules and knowledge of drug resistance pathways are also used to inform the classification of AMR mutations.

By contrast, GAM/ML analyses rely on computational methods to identify novel mutations that are associated with drug resistance (significantly differ in frequency in groups of resistant and susceptible specific drugs), without prior expert knowledge. GAM evaluates large datasets of genetic variants and identifies statistically significant associations with drug resistance, providing a more data-driven, unbiased approach to identifying resistance markers. This differs from the WHO catalogue, which is more focused on known mutations, often limited by the clinical samples available at the time of its compilation. This distinction is now reflected in the Discussion as follows:

(Page 19. Line 357-366) " GAM also does not rely on prior knowledge of genes, pathways, or mechanisms involved in drug resistance, enabling it to comprehensively analyze all variants present in a dataset. In contrast, the current WHO approach is specifically tailored to Mtb and relies on identifying resistance-associated mutations through a combination of expert-driven rules, masking strategies to exclude naturally occurring or neutral mutations, pathway knowledge derived from prior genetic studies to pinpoint relevant gene functions, and relaxed statistical thresholds to account for rare mutations. [35, 36] While these strategies provide valuable insights within the constrained scope of Mtb, they inherently limit broader applicability. The ability of GAM to operate independently of such predefined rules or assumptions allows it to provide a more flexible and data-driven framework for identifying drug resistance-associated variants across diverse pathogens."

[35] WHO. Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance. (2021).

[36] WHO. Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance second edition. (2023).

Response to Reviewer #3
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

**Response:** We appreciate your effort in improving our manuscript and have provided a point-to-point response in the listed reports.

Reviewer #3 (Remarks on code availability):
Code repository was not available for review.

**Response:** We regret that our code was not available for your review. We previously submitted our code to CodeOcean per the recommendation of Nature Communications. However, we experienced repeated technical difficulties with the CodeOcean platform, which frequently and spontaneously deleted our code at various intervals after uploading, requiring multiple re-uploads. Due to these reliability issues, we have discontinued using CodeOcean and have reverted to GitHub. The GitHub repository (github.com/jsaliba1/GAM) is now open and contains the code along with all relevant data files for this manuscript.

**RESPONSE TO REVIEWERS' COMMENTS**

Reviewer #2 (Remarks to the Author):

Comments to the authors:

The authors significantly improved their manuscript, making their work much more transparent and reproducible. Also, the code repository is now available and well structured, and data to run the analyses is provided.

Remaining comments

1) The full data for the GAM analysis is available, but in addition a table listing the 127 resistance groups as well as one with the assignments of the samples to the different groups would be helpful.

We appreciate your feedback and have now uploaded an excel file to Figshare (**S10**) that lists all the groups as well as displays the sample ID assigned to each group.

2) Any description of the LMM analysis is missing in the method section.

We are sorry for any confusion, the LMM analysis was not given its own section describing it in the methods, but it is fully described in the **Sample size and data completeness analyses** section of the methods.

The text reads "The LMM model employed in this analysis used FaST-LMM[58] with adjustments for kinship, lineage, geographic region of sample collection, and the testing site of each sample, where the kinship matrix was assessed using the Jaccard similarity index, and lineage, geographic region, and test site were included as covariates."

[58] Lippert, C., et al. FaST linear mixed models for genome-wide association studies. Nat Methods 8, 833-835 (2011).

3) Abbreviations in tables (e.g. SEN_lci in Supplementary Table 3) should be explained.

We have added a description to the table explaining the abbreviations.

"SEN and SPEC denote sensitivity and specificity, respectively, while UCI and LCI refer to the upper and lower confidence intervals, respectively."

4) Supplementary Table 4: Please add at least the TN numbers, potentially also PPV or accuracy, so that full performance metrics are available.

We are sorry for this oversight and have now added the TN number to **Supplementary Table 4** for both GAM and LMM.

**Supplementary Table 4. True positive (TP), false positive (FP), false negative (FN), and true negative (TN) associations detected by GAM and LMM in *S. aureus*.**

| | TP | | FP | | FN | | TN | |
|---|---|---|---|---|---|---|---|---|
| | *GAM* | *LMM* | *GAM* | *LMM* | *GAM* | *LMM* | *GAM* | *LMM* |
| **GEN** | 1 | 1 | 16 | 5958 | 0 | 0 | 420547 | 414605 |
| **PEN** | 0 | 2 | 17 | 110017 | 3 | 0 | 420544 | 310545 |
| **MET** | 1 | 1 | 257 | 140196 | 0 | 0 | 420306 | 280367 |
| **FUS** | 10 | 10 | 16 | 87253 | 12 | 10 | 420526 | 333291 |
| **TEI** | 0 | 0 | 0 | 1360 | 0 | 0 | 420564 | 419204 |
| **VAN** | 0 | 0 | 0 | 579 | 0 | 0 | 420564 | 419985 |
| **ERY** | 0 | 2 | 74 | 94352 | 4 | 0 | 420486 | 326210 |
| **CLI** | 0 | 1 | 115 | 4356 | 4 | 0 | 420445 | 416207 |
| **LIN** | 0 | 0 | 0 | 866 | 1 | 1 | 420563 | 419697 |
| **CIP** | 6 | 4 | 700 | 150224 | 7 | 7 | 419851 | 270329 |
| **RIF** | 1 | 13 | 0 | 1905 | 15 | 1 | 420548 | 418645 |
| **TET** | 2 | 2 | 30 | 41393 | 1 | 0 | 420531 | 379169 |
| **TMP** | 5 | 3 | 47 | 8661 | 2 | 5 | 420510 | 411895 |

5) For reproducibility, a short summary for the most important items in the supplied data folder on Figshare (e.g. content and dimensions, what would be input file, what is a result file, …) would be very helpful.

Thank you for your suggestion. We have edited the readme file in our GitHub repository to include a short summary that addresses this issue. The readme file now has a section titled Data File Summary that reads:

“**MUTATIONS.csv** - Contains mutation data for all isolates collected by the Cryptic consortium. This file is used as an input for GAM. UNIQUEID (string): Identifier for the isolate. MUTATION (string): Mutation label. GENE (string): Gene where the mutation occurs. IS_SYNONYMOUS (Boolean): Indicates if the mutation is synonymous.

**ENA_drug.csv** - Stores resistance phenotypes for different drugs. It is used to define resistance groupings in GAM. UNIQUEID (string): Identifier for the isolate Drug columns (binary: Resistant/Susceptible): Resistance status for each drug.

**bugs.csv** - Contains a list of isolates to be included in the study, ensuring they belong to a group relevant to the GAM analysis. ID (string): Identifier for the isolate. Group (string): Group assignment.

**DST_SAMPLES.csv** - Metadata file providing lineage, site, and country of origin for each sample. Used to categorize isolates in the analysis. UNIQUEID (string): Identifier for the isolate. LINEAGE (string): Lineage classification. SITE (string): Sampling site. COUNTRY (string): Country of origin.

**GAM_train.csv** - The main input file for the machine learning (ML) model, indicating the presence or absence of significant mutations identified by GAM. UNIQUEID (string): Identifier for the isolate. Mutation columns (binary: 1/0): Presence or absence of significant mutations.

**Y9_train.csv** - Label file for the training dataset used in the ML model. Corresponds to GAM_train.csv. UNIQUEID (string): Identifier for the isolate.

**LMM/** (Folder) - Stores results from the LMM analysis across different sample sizes. These results are used as input for calculating summary statistics and filtering significant mutations."

Reviewer #2 (Remarks on code availability):

Well structured.

Thank you for your positive comments on our code.

Reviewer #3 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Thank you for your effort and we have included the response along with the listed reports.